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August 7, 2025

Genevieve Johnson, Acting Regional Director
Lower Colorado Regional Office
U.S. Bureau of Reclamation
P.O. Box 61470
Boulder City, Nevada 89006

Dear Ms. Johnson:

Enclosed for your information is the New Mexico Interstate Stream Commission's Report for 2024 on irrigated acreage, diversion and consumptive use of water in the Gila River Basin in New Mexico. The records contained in the report are compiled pursuant to Article VII of the Consolidated Decree entered by the United States Supreme Court in *Arizona v. California*, 547 U.S. 150 (2006).

The Consolidated Decree limits the total consumptive use of water from each of three stream systems in New Mexico: the San Francisco River, Gila River (exclusive of Virden Valley) and San Simon Creek. Based on the consumptive use calculations prepared by the Interstate Stream Commission, the consumptive use for each stream system for 2024 was within the decree limitations on use. The decree also limits the total irrigated acreage within specified geographic areas of the Gila River Basin in New Mexico. Based on records from a field survey conducted by the New Mexico Office of the State Engineer during 2024, the acreage irrigated within each geographic area was within the decreed acreage for the respective areas.

Please call me at 505-469-9168 or email me at trevor.birt@ose.nm.gov if you have any questions regarding this information.

Sincerely,

A handwritten signature in blue ink that reads "Trevor A. Birt".

Trevor Birt – NMISC Hydrologist

**REPORT FOR 2024
ON IRRIGATED ACREAGE,
DIVERSION AND CONSUMPTIVE USE OF WATER
FOR ALL PURPOSES IN
NEW MEXICO FROM THE GILA RIVER,
SAN FRANCISCO RIVER
AND
SAN SIMON CREEK
(GILA RIVER BASIN IN NEW MEXICO)**

**Arizona v. California,
547 U.S. 150 (2006)**

**New Mexico Interstate Stream Commission
New Mexico Office of the State Engineer, District III**

INTRODUCTION

This report is prepared to provide annual records of irrigated acreage, surface and groundwater diversions and consumptive use of water within the Gila River Basin in New Mexico, which comprises the Gila River, San Francisco River and San Simon Creek, their tributaries and underground water sources. The records contained in this report are compiled pursuant to Article VII of the Consolidated Decree entered by the U.S. Supreme Court in *Arizona v. California*, 547 U.S. 150 (2006).

On the following pages, data for the year 2024 are summarized in Tables I through IV, including consumptive uses for the ten-year consecutive period.

Appendix I is a brief discussion of the methods and criteria used in determining the amount of irrigated acreage and in computing consumptive use by irrigation, stock, and reservoir and stock tank evaporation. Technical references are listed in Appendix II.

IRRIGATED ACREAGE

Personnel of the District III New Mexico Office of the State Engineer made the annual irrigated crop acreage field inventory of the nine areas of the Gila River Basin in the summer of 2024. The resulting maps were used to derive irrigation acreage statistics for this report. The mapped inventory data show the locations of irrigated acreage and by reference are made a part of this report and are available upon request.

The 2024 inventory found that the total acreage irrigated in each of the areas in 2024 was within the acreage limitation set forth in the Consolidated Decree. The water rights for irrigation have been adjudicated by State court decree, including those rights adjudicated to lands administered by the Forest Service. To the best of the Interstate Stream Commission's knowledge and belief, this report accurately depicts irrigation use on all acreage irrigated in the Gila River Basin that is subject to the Consolidated Decree.

CONSUMPTIVE USE

The consumptive use of water in 2024 in the Gila River Basin has been computed for the following categories: irrigation, stock, domestic, municipal, and industrial purposes, as well as reservoir and stock tank evaporation.

The total amount of consumptive use in 2024 for each of these categories, the total annual consumptive use in each stream system and the consumptive use limits decreed by the Supreme Court are listed in Table II. The consumptive uses in 2024 were within the annual limitations set forth in the Consolidated Decree.

The total amount of consumptive use in each stream system for the period of ten consecutive years, 2015-2024, was within the corresponding consumptive use limitation set forth in the Consolidated Decree. Please see Appendix I of this report for an explanation of the methods and criteria employed in computing consumptive uses.

Consumptive Use - Irrigation

The consumptive irrigation requirements (CIR) for the various areas were determined using the crop patterns found in the 2024 irrigated acreage inventory performed by the Office of the State Engineer and using gridded climate data (GridMET). The amount of water consumed incidental to the irrigation of crops is estimated to be 2% of the CIR in the San Simon and Virden Valley areas, where the lands under the Consolidated Decree are served primarily by pumps. Water consumed incidental to the irrigation of crops from groundwater in the Cliff-Gila and Redrock areas, partially served by wells, is also estimated at 2% of the CIR. Incidental amounts consumed by irrigation of acreage in the San Francisco River and Upper Gila areas, and by surface water irrigation in the Cliff-Gila and Redrock areas, are estimated to be 10% of the CIR.

Consumptive Use - Stock Tanks

The consumptive use from stock tanks is the evaporation loss from the exposed water surface area. The amount of water consumed from the tanks by livestock is tabulated as livestock use. Seepage from stock tank storage is not considered a consumptive use because it eventually returns to the groundwater system.

Consumptive Use - Livestock

Estimated consumptive use rates were applied to the reported livestock population in the four divisions of the Gila River Basin. A per capita use rate for cattle of 10 gallons per day was used in the San Francisco River region and the Gila River region upstream from Virden. In the Virden and San Simon areas, the per capita consumption rate used was 12 gallons per day (4, 5 Appendix II). The population of horses, hogs and other domestic animals is considered small, and a separate computation is not warranted. Their use is considered included within the use computed for cattle and sheep.

Consumptive Use - Municipal and Industrial

The consumptive uses for municipal purposes, including Silver City, are listed in Table II. Silver City diverts its water from the Gila River Basin to the Mimbres River Basin and its consumptive use is considered equal to the amount diverted from the Gila River Basin. Consumptive use by municipal and domestic water in the Basin is estimated as 50 percent of the metered diversion.

The use by Freeport-McMoRan Corporation (formerly Phelps Dodge Corporation) from its groundwater source near Tyrone is for industrial purposes. Water is also diverted by Freeport-McMoRan from the Gila River to Bill Evans Reservoir and hence to Tyrone for use in mining and milling operations. Freeport-McMoRan submits monthly reports to the Gila River Watermaster of their diversion, storage and use of Gila River water including groundwater usage and return flows. The reports include computation of the consumptive use of Gila River Basin water diverted by Freeport-McMoRan. Copies of the Freeport-McMoRan reports are on file in the District III Office of the State Engineer in Deming.

Consumptive Use - Domestic

Domestic uses are served primarily by municipal or domestic water systems and by “non-consumptive use wells”. Non-consumptive use wells are those that may provide water for in-house use only and such limited use is considered *de minimis*. The District III Office of the State Engineer in Deming reports municipal and domestic system diversions.

Consumptive Use – Lake Surface Evaporation

In 2024, four reservoirs were operated in the Gila River Basin. Lake Roberts, Snow Lake and Wall Lake are recreational reservoirs, and Bill Evans Lake serves to regulate water for industrial use and affords fishing and camping opportunities.

Records of water levels, average monthly water surface area, pan evaporation and precipitation are recorded and used to determine the evaporation loss from Bill Evans Reservoir. Pan evaporation rates and precipitation applicable to the three recreation reservoirs have been obtained by extrapolation from other locations. For evaporation calculations, it was assumed that Lake Roberts, Snow Lake and Wall Lake were maintained at full capacity throughout 2024.

DIVERSIONS

Table IV lists total diversions for 2024. The figures represent the sum of the measured diversions, as measured and/or reported by the District III Office of the State Engineer, the New Mexico Interstate Stream Commission, Freeport-McMoRan Tyrone, Inc. and Silver City, and unmeasured diversions, which are estimated by staff of the Interstate Stream Commission. Reported diversions are gross diversions including regulatory waste. Deliveries at points of use are not all measured.

Table I. Acreage Survey 2024

Area	Decreed Acreage <i>Arizona v. California</i>	Acres Irrigated
SAN FRANCISCO RIVER		
Luna	225	208
Apache-Aragon	316	162
Reserve	725	350
Glenwood	1,003	731
TOTAL	2,269	1,451
GILA RIVER exclusive of Virden Valley		
Upper Gila	287	85
Cliff-Gila	5,314	2,130
Redrock	1,456	116
TOTAL	7,057	2,331
GILA RIVER Virden Valley	381	60
SAN SIMON San Simon	2,900	755

**Table II. Annual Consumptive Use
2024 (Units: Acre-Feet)**

	AGRICULTURE										
Area	Irrigation	Stock Tank Evap	Stock	Municipal, Industrial & Domestic	Irrigated Land Acres	Crop Use Full Supply	Shortage to Full Supply	Incidental Use	Lake Surface Evaporation	Annual Use	Decreed Use
SAN FRANCISCO RIVER											
Luna	268				208	322	76	22			
Apache-Aragon	23				162	254	231	0			
Reserve	278				350	599	342	22			
Glenwood	1333				731	1683	459	108			
TOTAL	1901	423	156	65	1,451	2,858	1,109	152	0	2,546	4,112
GILA RIVER exclusive of Virden Valley											
Upper Gila	169				85	159	0	11			
Cliff-Gila	5511				2,130	5,038	0	473			
Redrock	227				116	337	131	21			
TOTAL¹	5908	1,154	160	6,449	2,331	5,534	131	505	562	14,233	15,895
GILA RIVER Virden Valley											
	180	30	15	9	60	169	0	11	0	234	1,103
SAN SIMON San Simon											
	934	128	17	9	755	934	0	0	0	1,089	8,220
¹Includes	5,683	Acre-feet of consumptive industrial use by Freeport-McMoRan and 663.33 acre-feet exported to the Mimbres Basin for municipal use by Silver City									

Table III. Consumptive Use for the Period of Ten Consecutive Years, 2014-2024 (Units: Acre-Feet)

Year	Stream System		
	San Francisco River	Gila River - exclusive of Virden Valley	San Simon Creek
2015	2,324	9,496	1,169
2016	2,461	11,286	1,170
2017	2,843	11,932	2,179
2018	2,309	12,446	2,606
2019	2,608	12,716	817
2020	3,268	11,032	1,523
2021	2,599	10,938	1,196
2022	1,877	10,577	807
2023	3,060	14,305	1,129
2024	2,546	14,233	1,089

Total 10 Consecutive Years 25,895 118,961 13,685

Max Decreed 10 Consecutive Years 31,870 136,620 72,000

Table IV. Diversions 2024 (Units: Acre-Feet)

Area	Diversions
San Francisco	13,822
Gila - exclusive of Virden Valley	41,017
Gila - Virden Valley	143
San Simon	3,570

APPENDIX I

Appendix I contains a discussion of methods and criteria used in determining the amount of irrigated acreage and the computations of consumptive use for irrigation, stock, and reservoir and stock tank evaporation.

Irrigated Acreage

Office of the State Engineer Deming District staff perform crop surveys of the Gila River Basin, San Francisco River Basin and San Simon Creek Basin during the summer months using geographic information system software on portable computers. The resulting maps provide parcel information including irrigated status and crop type from which associated acreage statistics are computed.

Consumptive Use - Irrigation

Consumptive use by crops is the unit amount of water used on a given area in transpiration, building of plant tissue, and evaporated from adjacent soil, water surface, snow, or intercepted precipitation in any specified time. The CIR is consumptive use less effective precipitation (1, Appendix II).

A determination of the consumptive irrigation requirement and the consumptive use of water by irrigated crops is made each year for the various areas in the Gila Basin using the method developed by Blaney-Criddle (1, Appendix II). This method takes into account the length of the irrigation season, the types of crops grown, the average monthly temperatures, the monthly percentages of yearly daytime hours and the effective precipitation during the irrigation season.

The irrigation season for each crop is that period when irrigation water is applied to beneficial use. For some crops this would be a part or all of the frost-free period plus a pre-frost and post-frost period. The frost-free period is that period each season between the last temperature of 32 degrees F or less in the spring and the first temperature of 32 degrees F or less in the fall. The effective precipitation is that part of total precipitation that is available for consumptive use. It is computed using the method set forth in Blaney and Hanson, New Mexico State Engineer Technical Report, No. 32, (3, Appendix II) and is also referred to as the U.S. Bureau of Reclamation method.

Data for the Blaney-Criddle consumptive use formula are obtained as follows:

1. Mean monthly temperature, frost dates, and total monthly precipitation are obtained from GridMET data (2, Appendix II).
2. Crop patterns are taken from the 2024 irrigated acreage inventory.
3. Crop coefficients (K) are taken from State Engineer Technical Report No. 32, (3, Appendix II).

The computed CIR is multiplied by the total number of irrigated acres to determine the consumptive irrigation use by crops in each area. The consumptive irrigation requirement computed under the procedure of this report represents a full water supply available at all times to the crops. Shortages to a full water supply are experienced in varying amounts in some areas. The data on the water supply available to irrigated lands is incomplete in some areas.

In the San Simon area, supplied by wells, there is generally a full supply of water. Due to supplements from wells, the Redrock area will have a full supply in most years and only in the extremely dry years will the available supply be insufficient. The Cliff-Gila area is similar to the Redrock area and only in the extremely dry years is there insufficient water to supply the

requirements of the irrigated lands served by surface water. The acreage irrigated from tributaries to the Gila River in the Cliff-Gila area may be in a short supply in certain years. However, the number of acres involved is quite small. Also, the Upper Gila area can be expected to have a full supply, except in the extremely dry years. The water supply in the San Francisco River is generally less than adequate to supply the needs of the irrigated lands except for areas near Glenwood. Some lands served by pumps have a full water supply.

Based on available flow or diversion data, the annual amount of shortage is determined as follows. Monthly diversion requirements for each metered ditch are computed based on the amount of acreage irrigated, the consumptive irrigation requirement, an irrigation efficiency factor, and a ditch conveyance efficiency factor. By comparing the computed monthly ditch diversion requirements to measured monthly diversions, monthly shortages to the diversion requirements are determined for each metered ditch. The resulting annual percentage shortage for all metered ditches is applied to the full-supply crop consumptive use to estimate the shorted consumptive use. For non-metered ditches that also are subject to water supply shortages, the percentage shortage to all metered ditches within the same irrigation area is applied to the full-supply crop consumptive use to estimate the shorted consumptive use.

In addition to the consumptive use by crops as determined under the procedure described above, there is a small amount of water consumed incidental to the irrigation of crops. These incidental losses include the evapotranspiration losses from the supply canals, certain return flow channels, and other specific areas that consume water because of the diversion of water. The total land and water area subject to these incidental losses is small. The lands served by wells have the smallest areas of incidental use. The incidental use has been estimated to be 2% of the consumptive irrigation requirement in the San Simon and Virden Valley areas, and on any lands served by pumps in the other areas. For the lands irrigated by surface water in the San Francisco River areas and in the Gila River areas exclusive of Virden Valley, the incidental use is estimated to be 10% of the consumptive irrigation requirement.

Consumptive Use - Livestock

The livestock population of the Gila River Basin is derived using the livestock inventory reported by the National Agricultural Statistics Service, which reports livestock population on a county basis. To obtain the livestock population in San Simon, the total area of the San Simon drainage was determined to be 7% of the total area of Hidalgo County and this percentage was applied to the reported cattle population. The same procedure was used for the Virden Valley area, this area being 6% of the total area of Hidalgo County.

The Gila River Basin drainage area comprises about 53% of the total area of Grant County and about 41% of the total area of Catron County. Applying these percentages to the reported livestock population for each county yields total consumptive use for livestock. The livestock figures for the Forest Service are calculated as well.

Consumptive Use - Reservoir Evaporation

A pan-to-lake coefficient of 0.70 is used to convert the pan evaporation rate to lake surface evaporation. The lake surface evaporation rate thus determined is an average annual value. The total annual precipitation is deducted from the average lake surface evaporation rate to arrive at a net annual evaporation rate at each reservoir location. The net evaporation rate is multiplied by the acres of water surface area to determine the evaporation loss at each reservoir. Lake surface evaporation is reported in Table II.

A pan-to-lake coefficient of 0.80 is used to convert the pan evaporation rate to surface evaporation for on-farm irrigation reservoirs. Evaporation from such reservoirs is included in irrigation use in Table II.

Consumptive Use - Stock Tanks

The computation of consumptive use from stock tanks considers only the evaporation loss from the exposed water surface area of the tanks. The hydrographic survey of the basin includes data on capacity and water surface area of all adjudicated stock tanks at spillway crest elevation. For numerous tanks in the basin, the hydrographic survey also determined the area within the tank delineated by prominent shoreline vegetation. For this report the line of prominent shoreline vegetation is considered to represent the average annual water surface elevation in the tanks. The data thus secured on these areas for all tanks was employed to determine the average annual water surface area, which is used in the equation to compute the annual evaporation loss from the lake surface of stock tanks.

Pan evaporation data in western New Mexico is extremely limited. Average annual pan evaporation rates for the several geographic areas in the Gila River Basin have been estimated using data from adjacent evaporation stations. The extrapolation of pan evaporation data included consideration of the temperature and elevation of the stations to which the extrapolation was made as compared to the same data for the station from which the record was secured. Annual variations in the rate of water surface evaporation cannot be determined because of insufficient data and the average rate is used. The annual precipitation recorded at the principal weather stations within each of the geographic areas is deducted from the average annual evaporation to determine the annual net evaporation rate.

A pan-to-lake coefficient of 0.80 is used to convert the estimated pan evaporation rate to stock tank water surface evaporation (4, Appendix II). The net evaporation rate times the average annual water surface area determines the evaporation loss in acre-feet for the average tank. The number of tanks in a given area is multiplied by this average evaporation loss rate per tank to arrive at the total evaporation loss from stock tanks.

An annual inventory of all stock tanks in the basin has not been deemed necessary. It is estimated that 15% of the adjudicated tanks will be out of service in any one year due to breached tanks, silted tanks and tanks that are dry for the entire year. The same principles are applied to compute evaporation of stock tanks within the National Forest. The values listed in Table II for stock tank evaporation reflect the correction for out-of-service tanks.

Diversions – Gila River Basin

The New Mexico Interstate Stream Commission gathers records of diversions by major irrigation canals in the Gila River Basin in New Mexico. The ditch measuring devices are generally located upstream of the first field turnout. Some ditches have wasteways below the point of measurement. Some ditches continually divert water when the river is at high stages, even though the water is not used for irrigation. Such diverted water is returned to the river through the wasteways. No measurements or estimates of wasteway returns to the river have been made. The Interstate Stream Commission estimates diversions by ditches that are so small that the cost of a measuring device is not warranted. The Office of the State Engineer collects diversion records for major irrigation wells that serve irrigated lands throughout the Gila River Basin, and also reports diversions in the Basin that serve the major municipal and industrial uses.

APPENDIX II

References

1. Blaney, Harry F., and Criddle, Wayne D., "Determining Consumptive Use and Irrigation Water Requirements," U.S. Department of Agriculture Technical Bulletin No. 1275, December 1962.
2. Abatzoglou, J. T. (2013), Development of gridded surface meteorological data for ecological applications and modelling. *Int. J. Climatol.*, 33: 121–131. Accessed <https://www.climatologylab.org/gridmet.html>.
3. Blaney, Harry F., and Hanson, Eldon G., "Consumptive Use and Water Requirements in New Mexico," New Mexico State Engineer Technical Report 32, 1965.
4. Pacific Southwest Inter-Agency Committee, "Stock Water Facilities Guide," March 1962.
5. Murray, Richard C., "Estimated Use of Water in the United States, 1965," U.S. Geological Survey Circular 556, 1968.